Lab Work – Function in Python

1. Railway Reservation Seat Analyzer

Problem Statement

A railway coach has seats represented as follows:

```
seats = [
  "Booked", "Available", "Booked", "Booked",
  "Available", "Available", "Booked", "Available",
  "Booked", "Booked", "Available", "Booked"
]
```

Requirements

Create the following functions:

1. count_seats(seats)

Returns the number of booked and available seats.

2. first_available(seats)

Returns the seat number of the first available seat.

3. occupancy_percentage(seats)

Returns the percentage of occupied seats.

4. display_available_seats(seats)

Displays all available seat numbers.

Sample Output

Booked Seats: 7
Available Seats: 5

First Available Seat: 2

Occupancy Percentage: 58.33%

Available Seat Numbers:
2 5 6 8 11

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2. Food Delivery Performance Tracker

Problem Statement

Delivery times (in minutes) for different orders are given below:

delivery_time = [28, 45, 60, 22, 35, 80, 40, 25, 55, 18]

Requirements

Create the following functions:

1. fastest_delivery(times)

Returns the shortest delivery time.

2. delayed_orders(times)

Returns a list of orders taking more than 45 minutes.

3. average_delivery_time(times)

Returns the average delivery time.

4. delivery_category(times)

Displays order categories:

- Fast → ≤ 30 minutes
- Normal → 31–45 minutes
- Delayed → > 45 minutes

Sample Output

Fastest Delivery: 18 minutes

Delayed Orders:
[60, 80, 55]

Average Delivery Time:
40.8 minutes

Categories:
28 -> Fast
45 -> Normal
60 -> Delayed
...

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3. Movie Review Sentiment Analyzer

Problem Statement

Movie reviews are stored as follows:

```
reviews = [
  "excellent movie",
  "average story",
  "excellent acting",
  "poor direction",
  "excellent visuals",
  "poor screenplay",
  "good music",
  "excellent climax",
  "average performance",
  "good cinematography"
]
```

Requirements

Create the following functions:

1. count_sentiments(reviews)

Counts:

- Excellent
- Good
- Average
- Poor reviews

2. most_common_word(reviews)

Returns the most frequently occurring word.

3. longest_review(reviews)

Returns the review containing the maximum number of characters.

4. reviews_with_keyword(reviews, keyword)

Displays all reviews containing a given keyword.

Sample Output

Excellent Reviews: 4
Good Reviews: 2
Average Reviews: 2
Poor Reviews: 2

Most Common Word:
excellent

Longest Review:
good cinematography

Reviews containing 'excellent':
excellent movie
excellent acting
excellent visuals
excellent climax